

MSTAR PRACTICE WORKSHEET**Lesson 6-14: The Distributive Property**

Grade 6 Mathematics · Standard 6.AT.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: A**Expand $6(n + 3)$ using the distributive property.

- ☒ **A** $6n + 18$
- ☐ **B** $6n + 3$
- ☐ **C** $6n + 9$
- ☐ **D** $n + 18$

$$6(n + 3) = 6 \times n + 6 \times 3 = 6n + 18.$$

Question 2 **SELECTED RESPONSE — CORRECT: B**Expand $4(5 - 2)$ using the distributive property.

- ☐ **A** 9
- ☒ **B** 12
- ☐ **C** $20 - 2$
- ☐ **D** 22

$$4(5 - 2) = 4 \times 5 - 4 \times 2 = 20 - 8 = 12.$$

Question 3 SELECTED RESPONSE — CORRECT: A

Expand $3(x + 4)$ using the distributive property.

- ☒ A $3x + 12$
- ☐ B $3x + 4$
- ☐ C $7x$
- ☐ D $x + 12$

$$3(x + 4) = 3 \times x + 3 \times 4 = 3x + 12.$$

Question 4 SELECTED RESPONSE — CORRECT: D

When using the distributive property with $a(b + c)$, you must multiply 'a' by:

- ☐ A Only b
- ☐ B Only c
- ☐ C $b + c$ first, then multiply
- ☒ D Both b and c

The distributive property says $a(b + c) = ab + ac$. You multiply the outside factor by EACH term inside the parentheses.

Question 5 SELECTED RESPONSE — CORRECT: A

Light Cue 1: Expand $3(x + 4)$ for the stage-light controller.

- ☒ A $3x + 12$
- ☐ B $3x + 4$
- ☐ C $3x + 7$
- ☐ D $x + 12$

$$\text{Distribute the 3 to each term: } 3 \times x + 3 \times 4 = 3x + 12.$$

Question 6

SELECTED RESPONSE — CORRECT: B

Light Cue 2: Expand $5(2 + n)$ to set two spotlight banks.

- ☐ A $7 + 5n$
- ☒ B $10 + 5n$
- ☐ C $10 + n$
- ☐ D $10 + 5$

Distribute the 5 to each term: $5 \times 2 + 5 \times n = 10 + 5n$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: B

Which expression is equivalent to $6(2x + 5)$?

- ☐ A $2x + 30$
- ☒ B $12x + 30$
- ☐ C $12x + 5$
- ☐ D $12x + 11$

The distributive property says $a(b + c) = ab + ac$. Multiply the outside factor by each term inside: $6 \cdot 2x = 12x$ and $6 \cdot 5 = 30$, so the expression is $12x + 30$.

- **A:** The 6 reached the 5 but not the $2x$. Distribute it to both terms inside.
- **C:** The 6 reached $2x$ but not the 5. Distribute it to BOTH terms inside.
- **D:** The 6 and the 5 were added rather than multiplied. Six times five is 30.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

- ☐ A Multiply the outside factor by the second term only and copy the first term: $2x + 30$.
- ☒ B Multiply the outside factor by every term inside the parentheses: $6 \cdot 2x = 12x$ and $6 \cdot 5 = 30$, giving $12x + 30$.
- ☐ C Multiply 6 by $2x$, then add 6 and 5 instead of multiplying them: $12x + 11$.
- ☐ D Multiply the outside factor by the first term only and copy the second term: $12x + 5$.

Expanding requires the outside factor to reach every term inside the parentheses. Skipping a term or adding instead of multiplying produces an expression that is not equivalent.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Pulling Out a Smaller Factor

Elena was asked to factor $18m + 24$ completely. She wrote $18m + 24 = 2(9m + 12)$ and said she was finished because 2 divides both terms.

Explain Elena's misconception and give the completely factored expression.

Model answer: Elena pulled out a common factor instead of the greatest common factor. Her expression is equivalent to the original, but $9m$ and 12 still share the factor 3. The GCF of 18 and 24 is 6, so $18m + 24 = 6(3m + 4)$. Checking by expanding: $6 \cdot 3m = 18m$ and $6 \cdot 4 = 24$.

2 points	Student explains that 2 is a common factor but not the greatest common factor of 18 and 24, so the expression is not completely factored, and gives $6(3m + 4)$.
1 point	Student gives $6(3m + 4)$ but does not explain that Elena used a common factor instead of the greatest common factor.
0 points	Student agrees that $2(9m + 12)$ is completely factored or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

Use the distributive property to explain why 6×98 can be calculated as $6 \times 100 - 6 \times 2 = 588$. Then create your own example of using the distributive property to make multiplication easier.

Model answer: Because $98 = 100 - 2$, you can rewrite 6×98 as $6(100 - 2)$ and distribute: $6 \times 100 - 6 \times 2 = 600 - 12 = 588$. The distributive property is what makes that rewrite legal, and it turns one hard multiplication into two easy ones — the same trick works for problems like 7×99 or 4×52 .

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.